

Supplementary Information for:

Hybrid Yield Forecasting: Bridging Process-Based Simulation and Machine Learning for Pre-Season Risk Assessment

Tim Schweitzer

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Abstract

This document provides technical details, comprehensive data specifications, model hyperparameters, and regional forensic analyses supporting the main text. This material ensures the reproducibility of the Expert System-Guided Ensemble pipeline and provides evidence for the regime-switching hypothesis.

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S1 Master Data Table

The operational forecasting pipeline integrates multi-source data streams to bridge the gap between seasonal climate precursors and field-level physiological responses.

Table S1: Comprehensive Data Sources and Specifications

Dataset	Source	Resolution	Role
Yield Data	Thuringian State Inst. / Destatis [Statistisches Bundesamt (Destatis), 2024]	District (Annual)	Target Variable
AgERA5	ECMWF / Copernicus [Service, 2024]	0.1° (Daily)	Historical Weather
SEAS5	ECMWF [Johnson et al., 2019]	1.0° (Monthly)	Seasonal Forecast
ERA5-Land	ECMWF [Muñoz-Sabater et al., 2021]	0.1° (Monthly)	Soil Moisture / Temp
SoilGrids	ISRIC [Hengl et al., 2017]	250m	Static Soil Properties
CropMap	DLR [DLR (German Aerospace Center), 2023]	10m	Spatial Mask
MOD13Q1	NASA LP DAAC [Didan, 2015]	250m (16-day)	Vegetation Anomalies
Teleconnections	NOAA (NAO, SCAND, MEI)	Monthly	Climate Indices

S2 Extended Mechanistic Formulas

The **Compound Failure Index** identifies yield crashes by aggregating stress vectors derived from WOFOST internal moisture balances [Boogaard et al., 1998] and SEAS5 anomalies.

M_3 Bio-Physical Limits Logic: The M_3 model applies a hierarchical decision tree to adjust the trend baseline (Y_{trend}):

1. **Water Crash:** If $Z_{water} < -0.95$:

$$M_3 = Y_{trend} \times (1.0 - (0.15 + |Z_{water} + 0.95| \times 0.25))$$

2. **Bumper Potential:** If $Z_{water} > 1.0$ AND $Z_{solar} > -1.0$:

$$M_3 = Y_{trend} \times (1.0 + (0.05 + (Z_{water} - 1.0) \times 0.05))$$

3. **Else:** $M_3 = Y_{trend}$

Scorch Index (Heat + Drought interaction):

$$\text{Scorch} = \max(Z_{\text{heat}} \times (-Z_{\text{bal}})^+, (Z_{\text{heat}} - 1.5)^+ \times 2.0) \quad (\text{S1})$$

Drown Index (Anoxia/Waterlogging):

$$\text{Drown} = (Z_{\text{anoxia}} - 0.8)^+ \times 2.0 \quad (\text{S2})$$

Late Start Index (Sowing delay):

$$\text{LateStart} = (Z_{\text{sow}} - 1.5)^+ \times 2.0 \quad (\text{S3})$$

Skill Score: The skill score quantifies the improvement of the ensemble model relative to the statistical trend baseline:

$$\text{Skill Score} = \left(1 - \frac{\text{MAE}_{\text{Ensemble}}}{\text{MAE}_{\text{Trend}}}\right) \times 100 \quad (\text{S4})$$

S3 Comprehensive Hyperparameters

Hyperparameters for the Expert System were configured following best practices for robust generalization in high-noise agricultural environments [Paudel et al., 2022].

Hyperparameters for S_{hybrid} were selected via a time-aware random search over 200 iterations, optimizing MAE on the walk-forward validation folds, following the gradient boosting architecture of Shahhosseini et al. [2021]. The final values reflect the best-performing configuration within the constrained search space.

Table S2: Hyperparameters for Sub-Models and Expert System

Model	Configuration
M_1 Trend	LinearGAM ($n_splines = 10$), ARIMA(1, 0, 0)
M_2 Robust	HuberRegressor: $\epsilon = 1.35, max_iter = 200$
M_3 Bio-Physical Limits	Solar-Gated Logic (Thresholds: $Z_{water} < -0.95, Z_{solar} > -1.0$)
S_{hybrid} Hybrid	XGBoost: $n_est = 1242, lr = 0.057, max_depth = 3$
Expert System	Thresholds: Safety ($S < 0.8$), Bumper ($S > 1.0$), Weights (0.6/0.4)

S4 Implementation Details

As discussed in Section 2.4 of the main text, the S_{hybrid} component targets the yield-to-trend ratio rather than absolute yield to improve local optimization convergence.

Yield-to-Ratio Scaling (S_{hybrid}):

```
# Predicting the Ratio between observed yield and WOFOST simulation
df['yield_ratio'] = df['actual_yield'] / df['stage1_forecast']
df['yield_ratio'] = df['yield_ratio'].clip(0.5, 1.5)
```

References

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